

76_Unlocking the Potential of Plasma-Circulating miRNAs in Early Oncological Diagnosis: From Traditional Assays to Next-Generation Sequencing Platforms"

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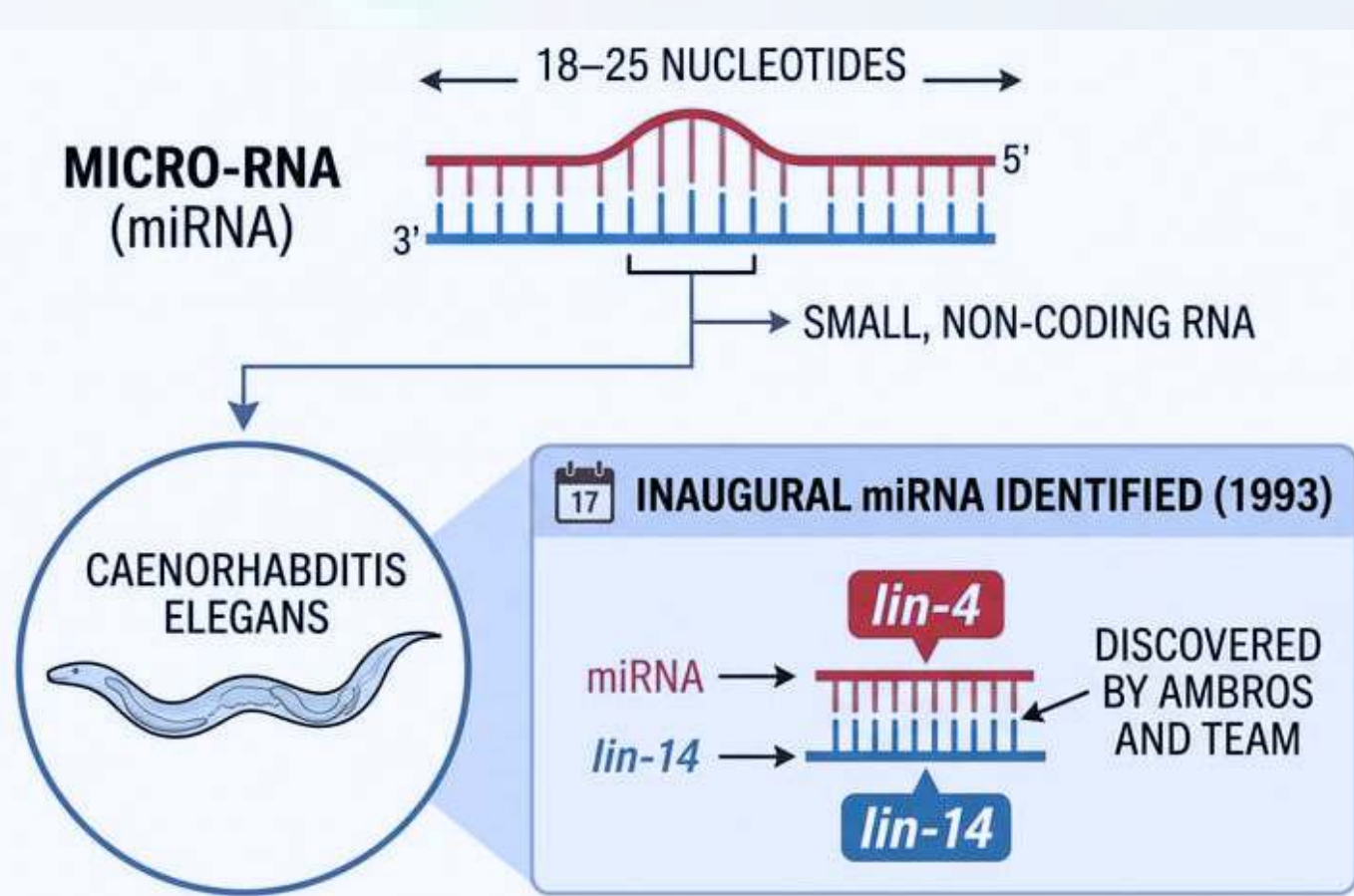
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This review focuses on the clinical application of microRNAs (miRNAs) as pivotal biochemical biomarkers to enhance the early diagnosis of asymptomatic and hidden malignancies, ultimately advancing precision medicine and patient outcomes. In this context, prior literature has highlighted significant preliminary obstacles to identifying distinct miRNA signatures specific to early-stage and challenging-to-detect tumors. To address this, this work comprehensively evaluates recent advancements in theoretical and clinical approaches, focusing on plasma-circulating miRNA panels optimized for the early screening and prognosis of highly aggressive cancers, such as non-small-cell lung cancer (NSCLC) and hepatocellular carcinoma (HCC). By examining the complex biochemical alterations in circulating nucleotide profiles, this study highlights how specific microenvironmental changes reflect early tumor development. Ultimately, while diagnosing early-stage tumors remains a clinical hurdle due to their asymptomatic nature, miRNAs continue to show remarkable promise across various malignancies. Emerging advanced high-throughput technologies, including target-specific digital profiling, novel biosensing platforms, and Next Generation RNA Sequencing (RNAseq), now significantly surpass traditional RT-qPCR standards in both sensitivity and specificity. These innovative methodologies offer advanced multiplexing capabilities and enhanced limit-of-detection (LOD) rates, thereby overcoming the historical heterogeneity of cancer types. Consequently, the integration of these molecular tools drives the future of clinical biochemistry, establishing robust diagnostic frameworks that pave the way for early therapeutic interventions.

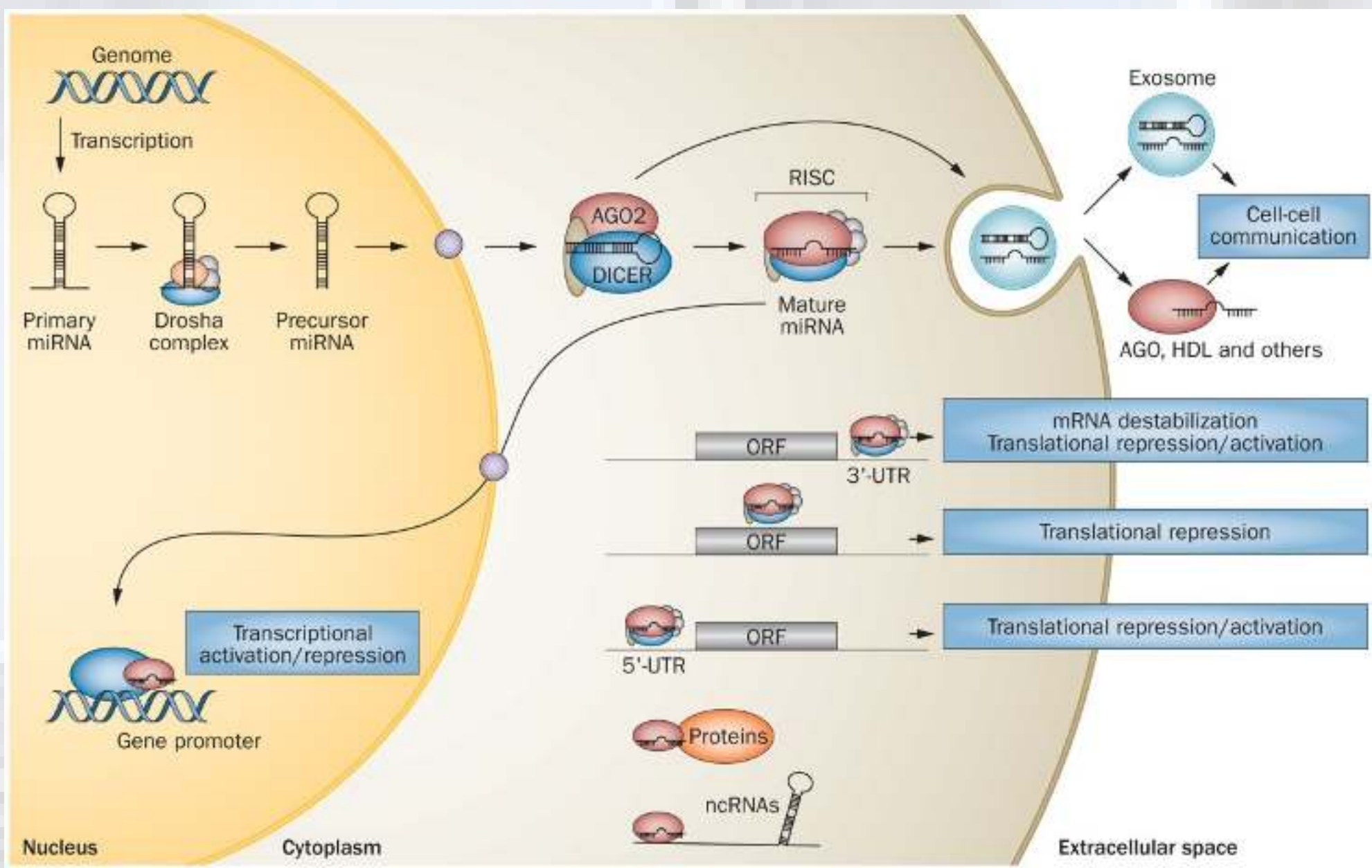
Keywords: miRNAs, Cancer, Biochemical Biomarkers, Clinical Approaches

What is miRNAs?

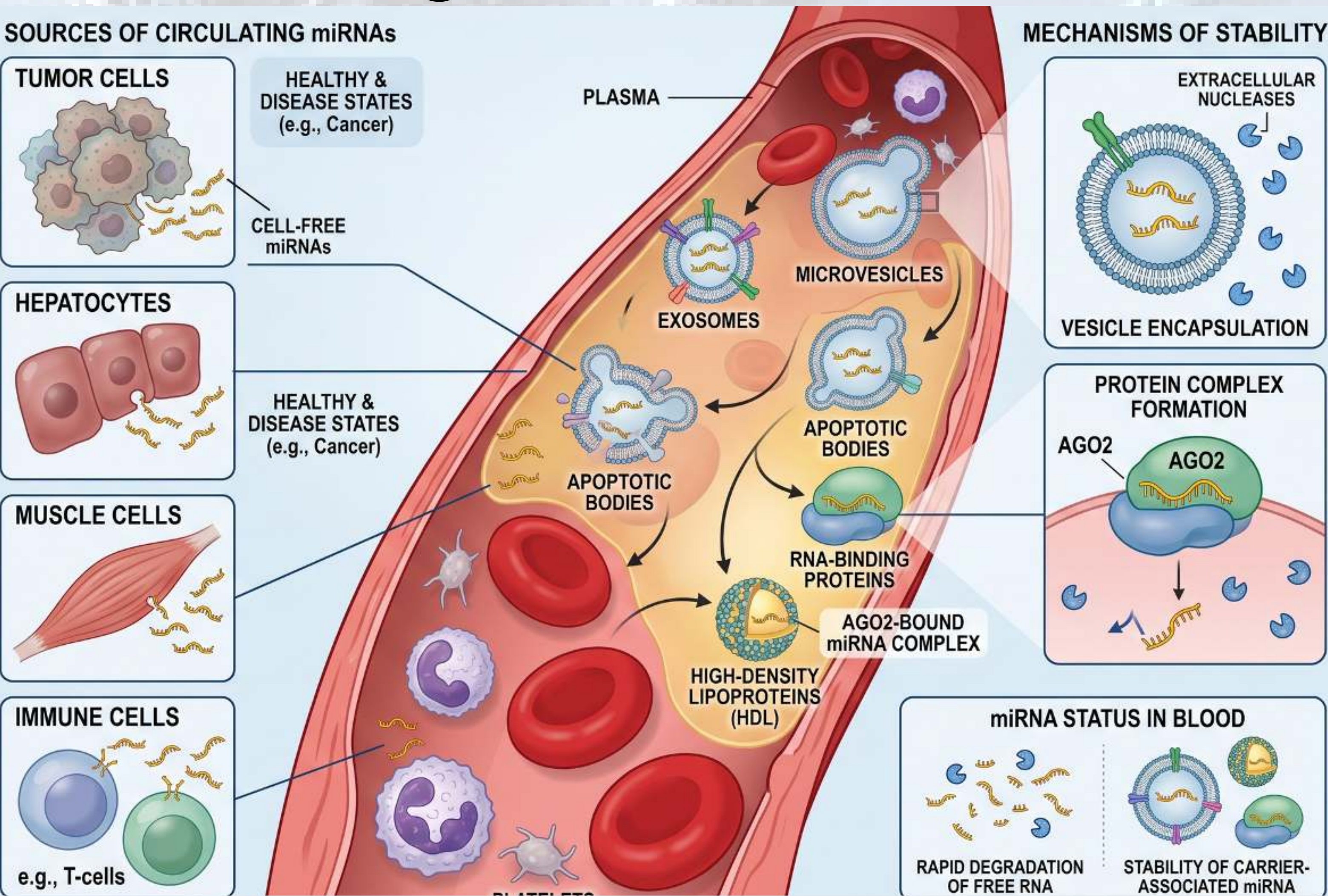
MicroRNAs (miRNAs) are small, non-coding RNA segments usually consisting of 18–25 nucleotides. The inaugural miRNA, *lin-4*, was identified in *Caenorhabditis elegans* by Ambros and his team in 1993



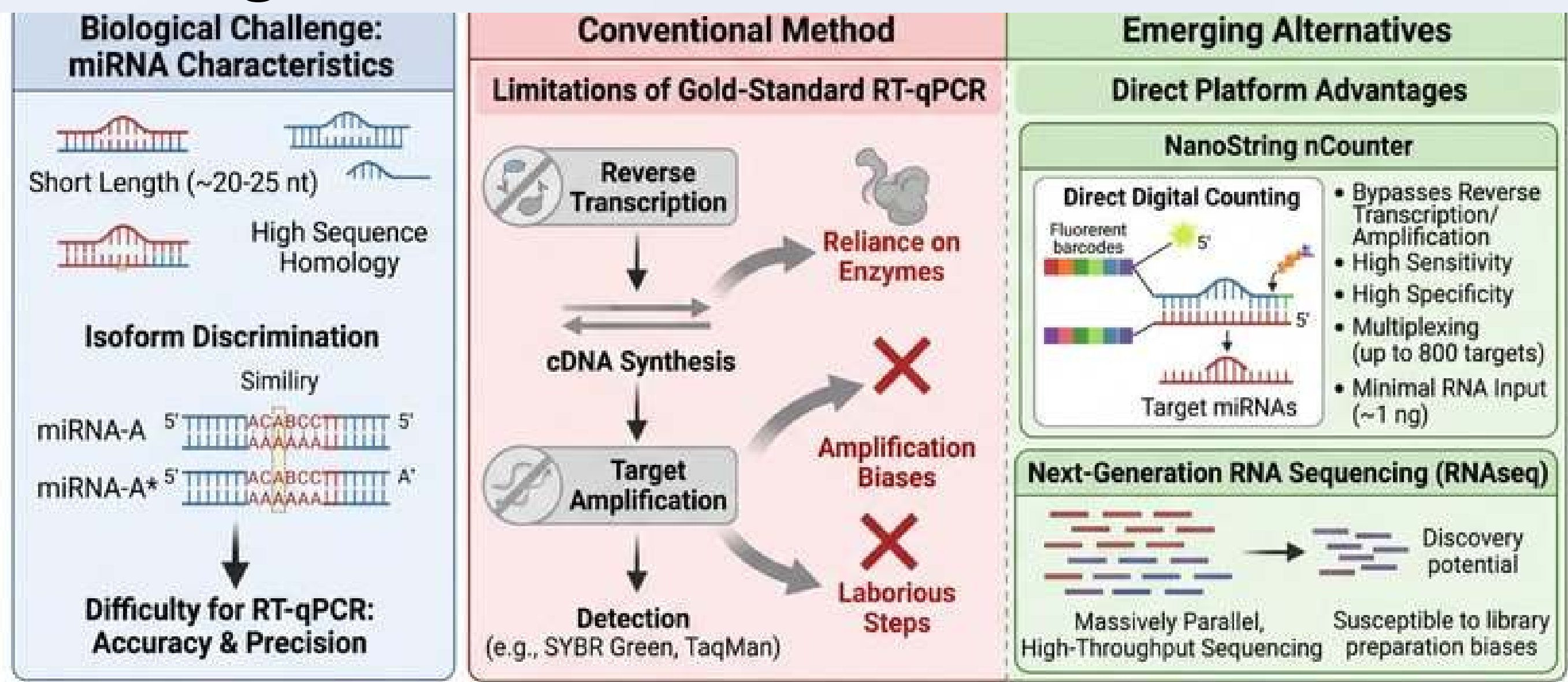
miRNAs excretion mechanism



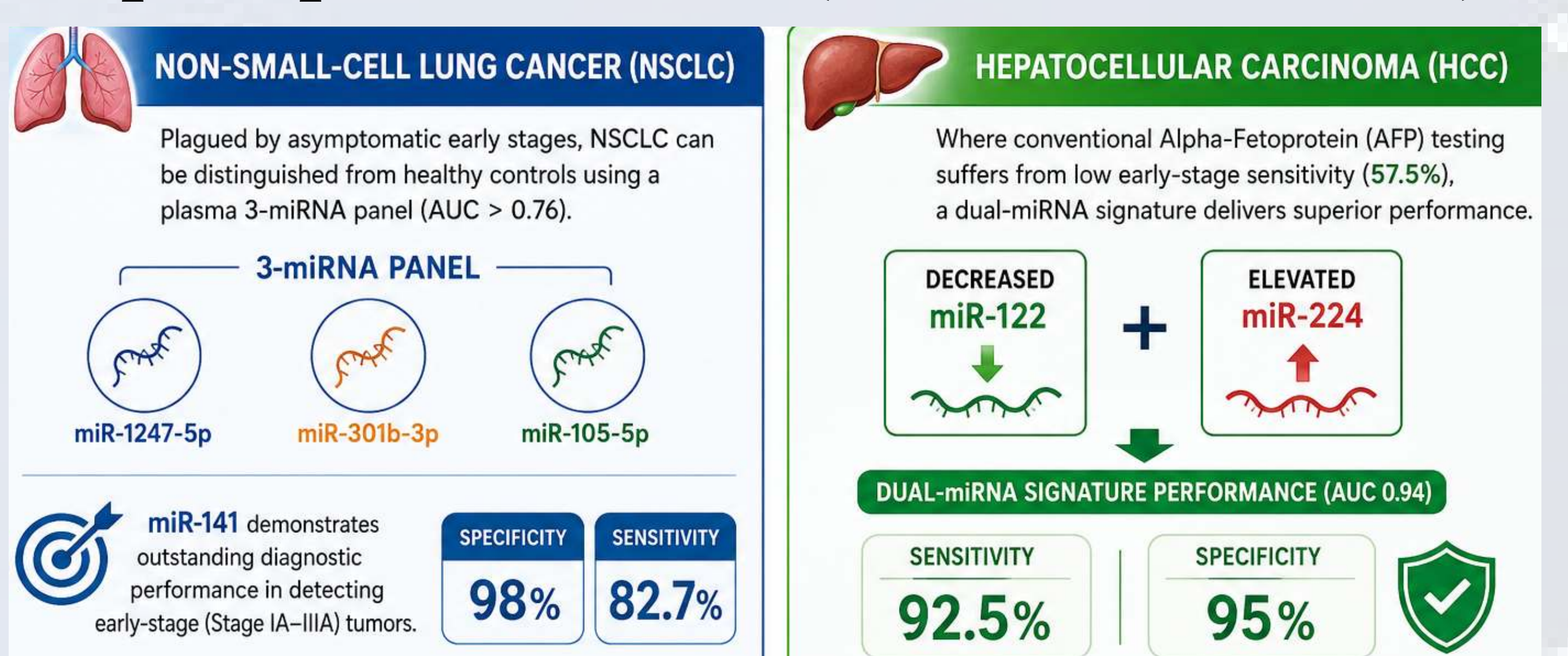
Circulating miRNAs



Technological Biosensor Advances for miRNA Detection



Clinical Application: Biomarker Signatures In Imperceptible Cancers (NSCLC and HCC)



Limitations of Circulating miRNAs

Normalization Deficit: No universal internal reference control exists for data normalization.

Hemolysis Interference: Red blood cell lysis during sampling inflates background miRNA levels.

Limited Tumor Specificity: Expression overlaps with systemic inflammation and benign conditions.

Conclusion

Circulating miRNAs hold significant promise as non-invasive biomarkers for early cancer detection. However, standardizing sample handling protocols, optimization of extraction procedures, and rigorous clinical validation remain essential to translate these biomarkers into routine clinical practice.

References

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